BULE HORA UNIVERSITY

College of Health and Medical Sciences School of Pharmacy

MEDICINAL CHEMISTRY GROUP PRESENTATION

LOCAL AND GENERAL ANESTHESIA

PREPARED BY

Pharmacy 2nd year GROUP 6 students

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OBJECTIVE OF PRESENTATION

❖**INTODUCTION==PART 1==**

- ✓We defined the terms anesthesia and anesthetics.
- ✓Type of anesthetic was listed and explained

- **LOCAL ANESTHETICS**
  - ✓Define well about local anesthetics
  - ✓History of local anesthetics described

OBJECTIVE OF PRESENTATION

- **CLASSIFICATION OF LOCAL ANESTHETIC BASED ON THE BASIS OF THEIR CHEMICAL STRUCTURES' part 2**
  - ✓ Properties of an ideal local anesthetic explained
  - ✓ SAR(structure activity of relation ship) of local anesthetics
- **GENERAL ANESTHETICS==part 3==**
  - ✓ Classification of General Anesthetic
  - ✓ Stages of General Anesthetic listed and explained
  - ✓ Some characteristics of ideal general Anesthetic

INTRODUCTION

WHAT IS ANSTHESIA ?

- **Anesthesia**, defined as a loss of sensation with or without loss of consciousness.
- **Anesthesia** is a treatment using drug called Anesthetics. These drugs keep patients feeling from pain during medical procedures.

Cont…

- **Anesthetic** compounds includes not only the classic anesthetic agents, such as the general and local anesthetics, but also many CNS depressants such as
  - analgesics
  - **sedative/hypnotics (barbiturates and (benzodiazepines)**,
  - Anticonvulsants and
  - skeletal muscle relaxants.
- Many of these agents are routinely used today in clinical practice to facilitate **surgical** and **medical** procedures.

TYPES OF ANESTHETIC AGENTS

- Generally, anesthetic agents are classified as
- **In Local anesthetics**

  The patients is **wide awake** during surgery to specific a small area.

- **In General anesthetics**

  The patients is **unconscious** and feeling nothing during surgery and patients receive medicine through **inhaled** or **IV** .

LET US SEE ONE BY ONE AS FOLLOWING





LOCAL ANESTHETICS



LOCAL ANESTHETIC AGENTS

- **A local anesthetic agents** are drugs that produced loss of sensation in **restricted area** of the body.
- Its Block the nerve that carries the pain sensation and automatic impulses in local areas of the body
- **Its Prevent conduction and formation of an action potential by either full or partial blockage of sodium ion channel.**
- **Its play important role in dentistry, ophthalmology and minor surgical operations.**

History of Local Anesthetic

- **As early as 1532**, the anesthetic properties of coca leaves (**Erythroxylon cocca**) became known to Europeans from the natives of **Peru**, who chewed the leaves for a general feeling of well-being and to reduce hunger.

- **The active principle** of the coca leaf, however, was not discovered until 1860 by **NIEMANN**, who obtained a crystalline alkaloid from the leaves, to which he gave the name **cocaine**, after that cocaine become used as local anesthetics.

CONT…

COCAINE (drug)



LEAVES OF COCCA(crude drug)



ERYTHROXYLON COCCA (genus-**species**)



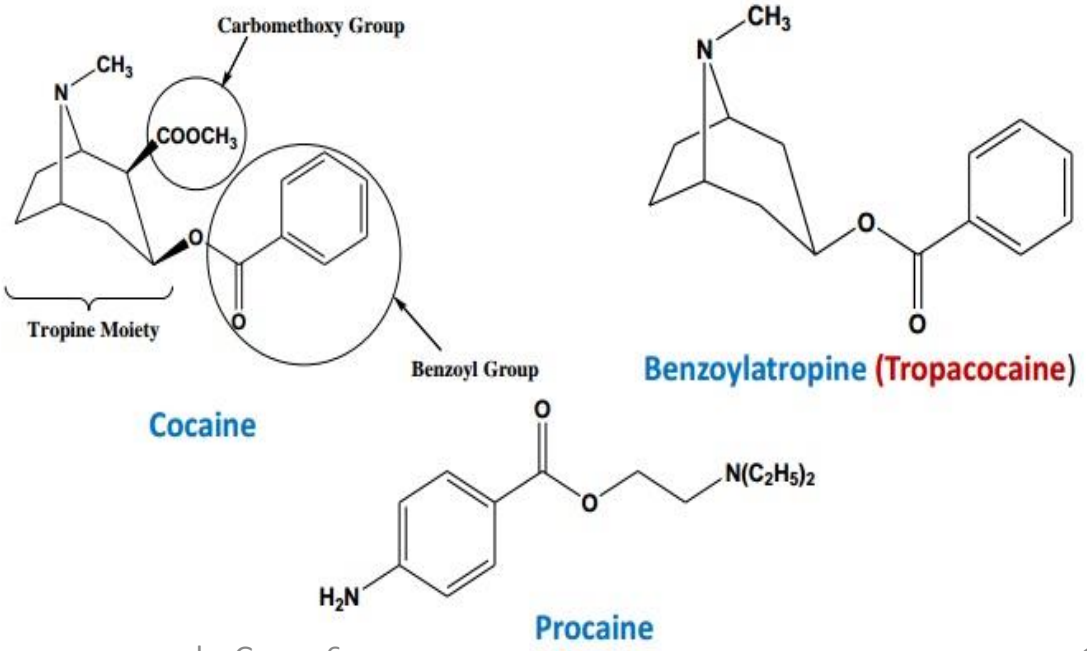
ERYTHROXYLACEAE (**family**)

CONT…

- The active principle---**cocaine**
  - Found to have **anesthetic effect**.
- Many attempts were made to prepare new analogues of cocaine that lacked its
  - **addicting liability**
  - **allergic reaction**
  - **poor stability in aqueous solution.**
- It was soon realized that a variety of benzoyl ester of amino alcohols, including **benzoyl atropine,** exhibit strong local anesthetic property without any addicting liability.

CONT…

- Thus, the removal of 2-carbomethoxy group of **cocaine** also abolished **addictive liability**
- **This discovery eventually led to the synthesis of procaine**.

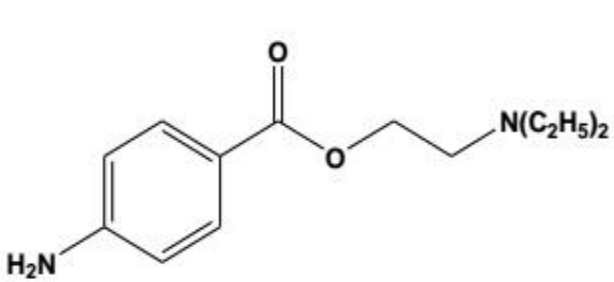


Procaine

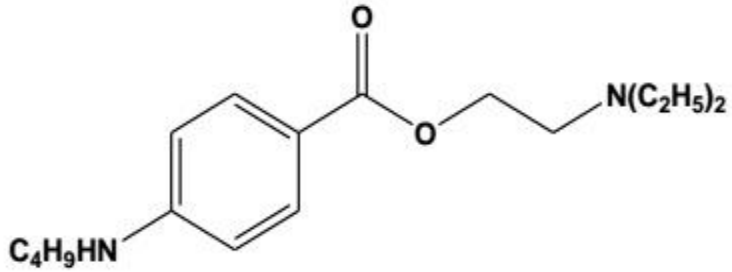
- Become a **prototype** for local anesthetics largely because it did not have the severe local and systemic toxicity as **cocaine**.
- The intrinsic potency of **procaine** was low and its duration of action was relatively short as compared to **cocaine .**
- But if it is combined with a vasoconstrictor such as **epinephrine** can be long.
  - **vasoconstrictor** agent reduces blood supply and there by prolonging the **residence time** of procaine.

CONT…

- **Various compounds were synthesized for the purpose of enhancing the intrinsic potency and the duration of action of procaine.**
- Among these compounds, ***tetracaine*** still is the most potent, long acting ester type LA used in **spinal anesthesia.**



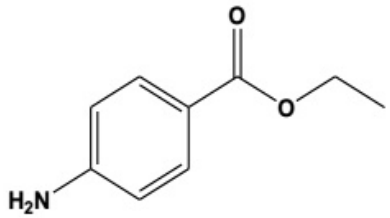
Procaine



Tetracaine

Benzocaine

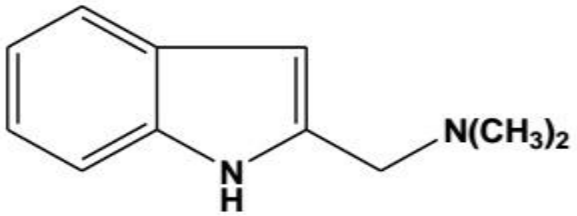
- **Benzocaine** was synthesized and found to have good anesthetizing properties and low toxicity (**topical anesthetic**)
- But **benzocaine** has limited water solubility except at low pH values owning to the lack of **a basic aliphatic amino group**
- There by disallowing the preparation of pharmaceutically acceptable parenteral solutions.



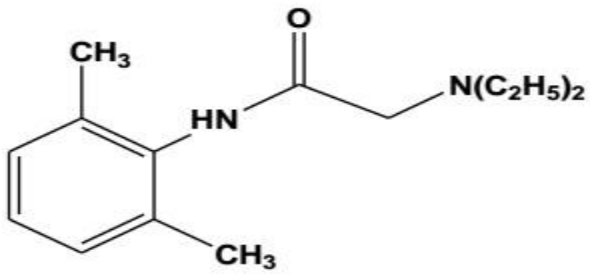
Benzocaine

CONT…

- The next major turning point in the development of LA was serendipitous discovery of local anesthetic activity of **Isogramine**.
- This observation led to the synthesis of **lidocaine**.



Isogramine



Lidocaine

Lidocaine

- The first non- irritating amide type local anesthetic agent.
  - Less suffer to allergic action than **procaine**
  - Stability in aqueous solution Because of the more stable amide functionality
- Structurally, **lidocaine** can be viewed as the open chain analogue of **isogramine**.

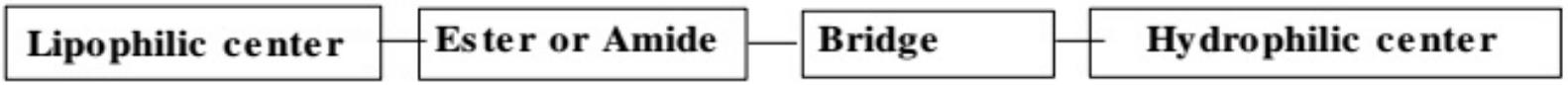
WELL COME TO

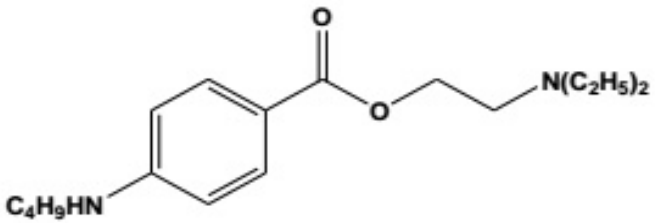
==PART==2

CLASSIFICATION OF LOCAL ANESTHETC

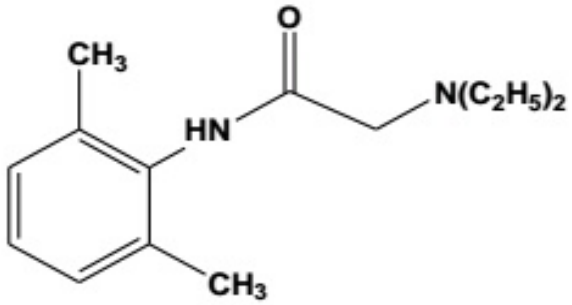
CLASSIFICATION OF LOCAL ANESTHETC

- Based on the their ‘chemical structures’
- **Ester type**
- **Amide type**





ester type



amide type

CLASSIFICATION OF LOCAL ANESTHETICS

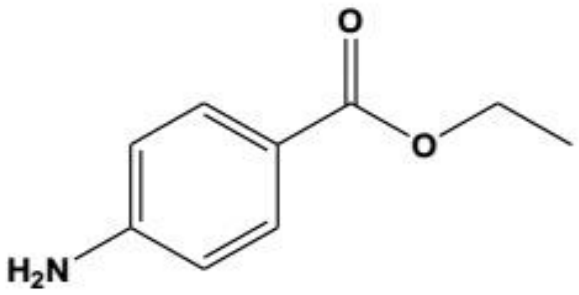
ESTER TYPE

- **Benzocaine**
- **Tetracaine**
- **Propracaine**
- **Procaine**

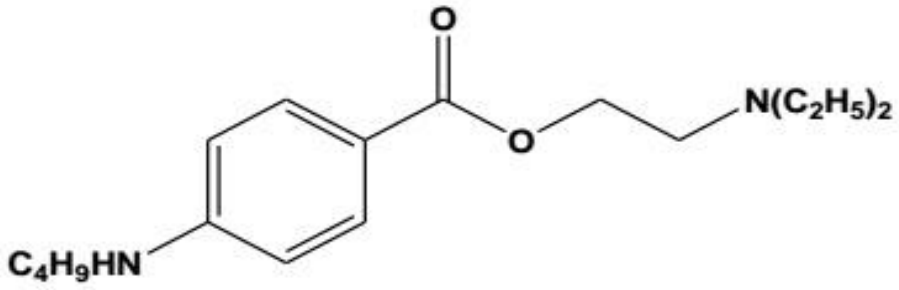
AMIDE TYPE

- Lindocaine
- Tolycaine
- Tocainide
- Prilocaine
- Mepivacaine
- bupivacaine

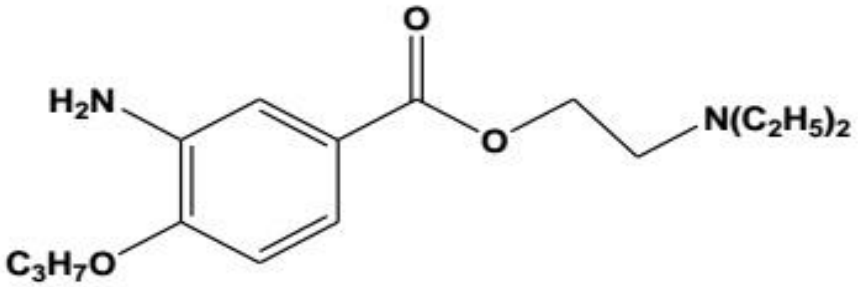
Ester type



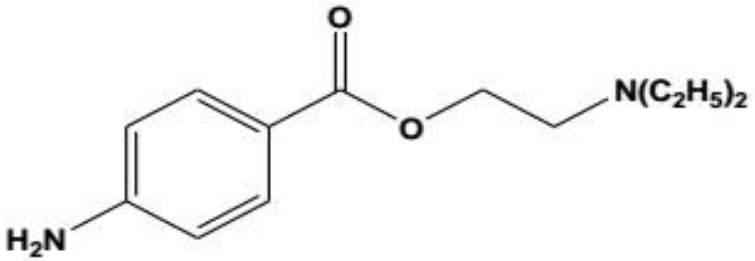
Benzocaine



Tetracaine

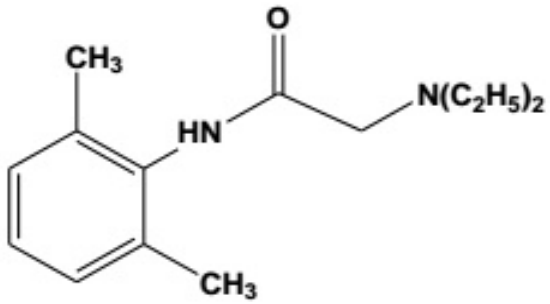


Proparacaine

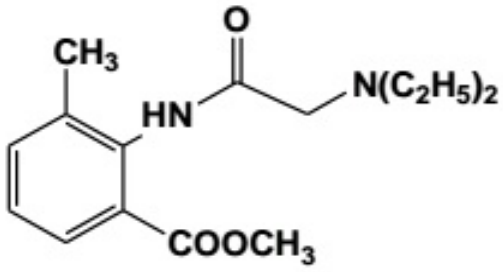


Procaine

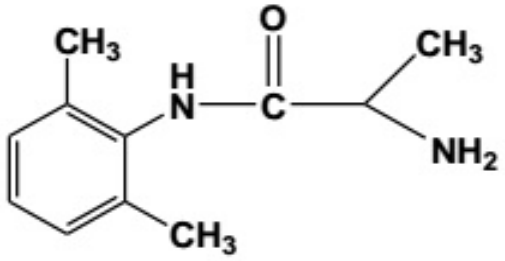
Amide type



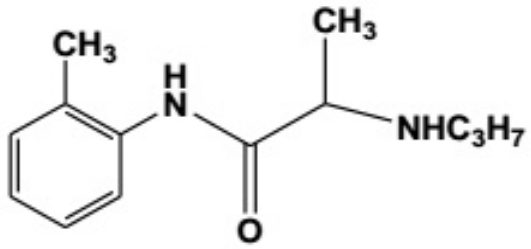
Lidocaine



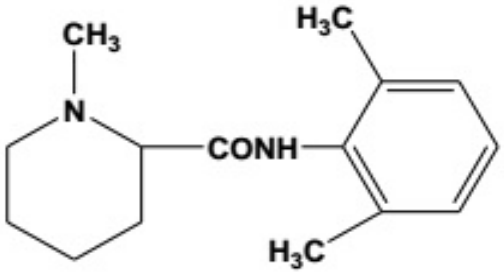
Tolycaine



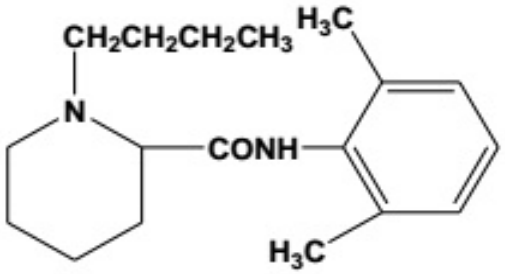
Tocainide



Prilocaine



Mepivacaine

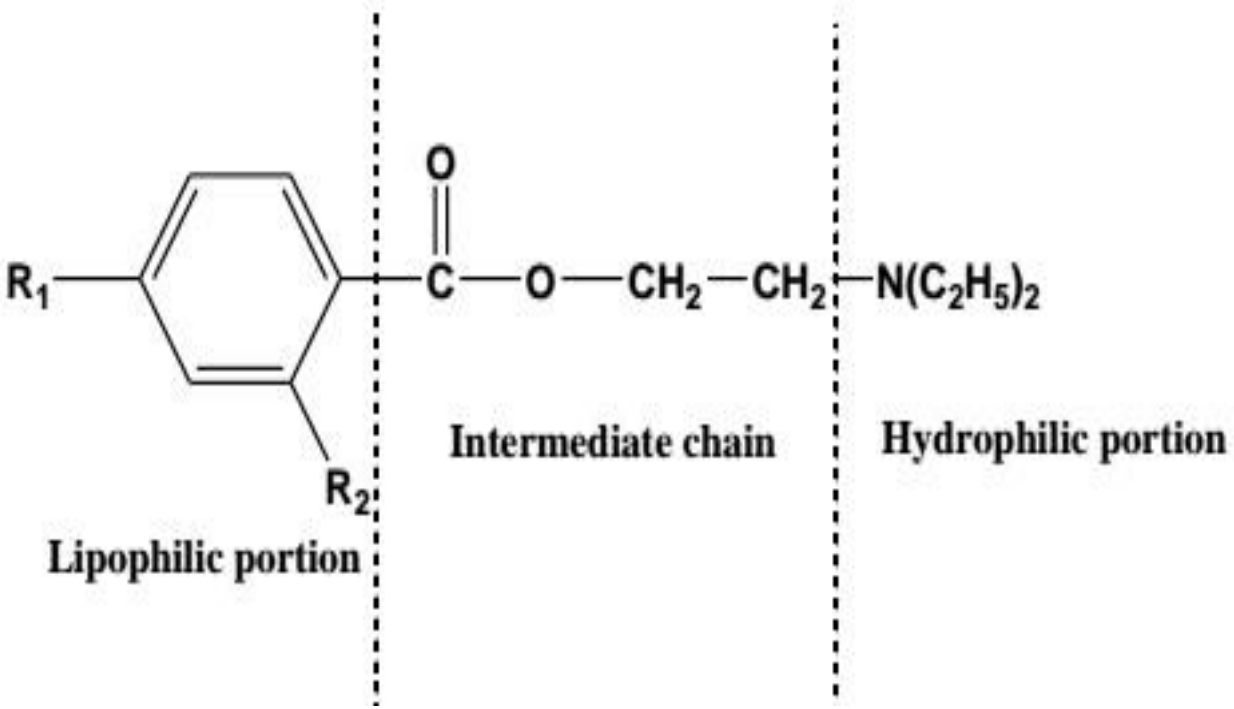


Bupivacaine

PROPERTIES OF AN IDEAL **LOCAL ANESTHETICS**

- **The ideal properties of local anesthetics are:-**
  - **The action of local anesthetic must be reversible blockade of nerve conduction**
  - **It should have a low degree of systemic toxicity**
  - **It should have sufficient potency to provide complete anesthesia**
  - **It should have rapid onset and be of sufficient duration to complete surgical anesthetic**
  - **It should be relatively free from producing allergic reactions**
  - **It should be either sterile or capable of being sterilized by heat without deterioration**

SAR



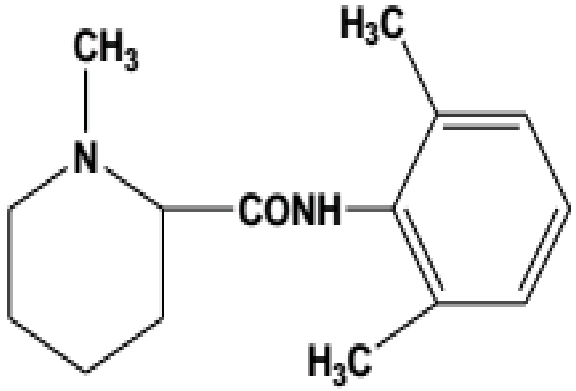
- **Local Anesthetics** have chemical structure that normally have the following general format.
  - Lipophilic portion
  - Intermediate chain
  - Hydrophilic portion

LIPOPHILIC PORTION

- **The lipophilic portion of the molecule is essential for local anesthetic activity.**
- **This portion of the molecule consists of either**
  - ✓ **An aromatic group directly attached to carbonyl function (the amino ester series)**

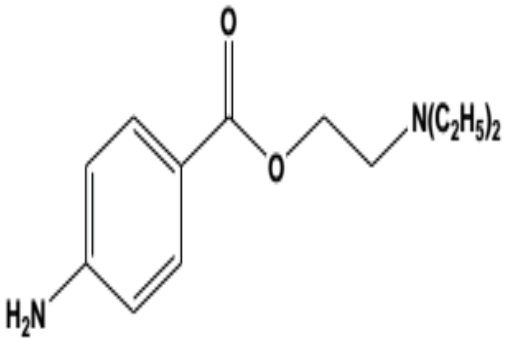
OR

  - ✓ **A 2,6-dimethylphenyl group attached to a carbonyl function through an —NH— group (the amino amide series).**



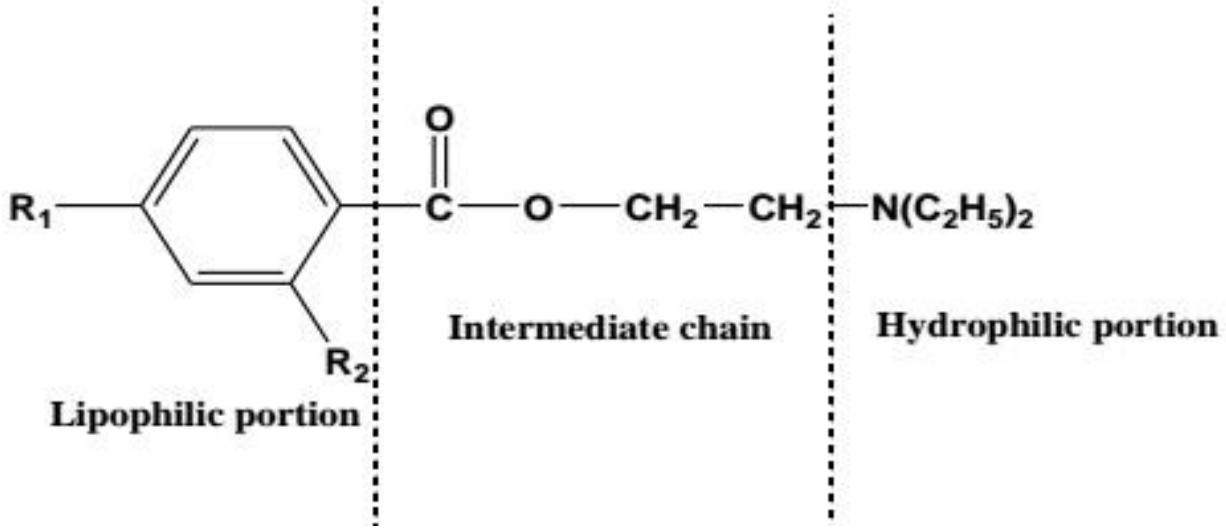
Mepivacaine

- **Both of those groups are highly lipophilic and appear to play an important role in the binding of LA to channel proteins. Eg. Mepivacaine**
- **In the amino ester series, an electro donating substituent in the ortho or para or both**
- **contribute electron density to the aromatic ring by both resonance and inductive effect**
- **enhancing LA potency over non substituted analogs**



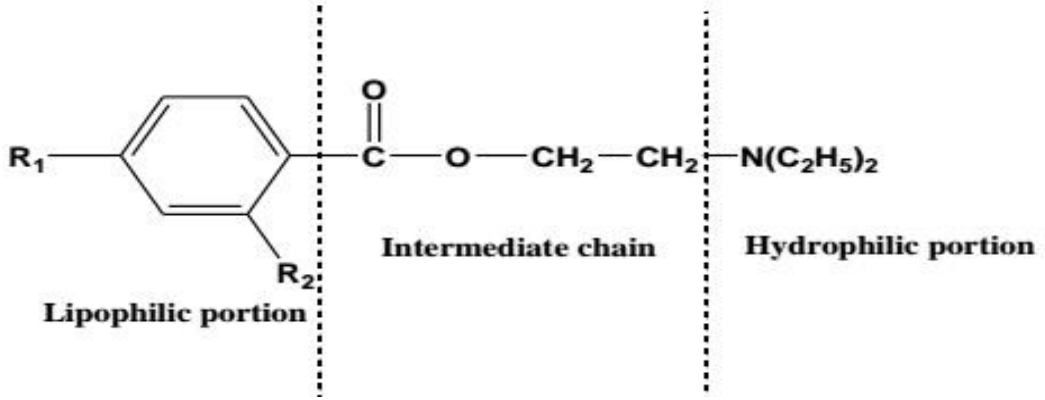
- Insertion of a methylene group between the aromatic moiety and the carbonyl function in procaine molecule.
  - Prohibit the formation of the resonance form
  - Led to procaine analog with greatly reduced potency
- When an amino or alkoxy group is attached to meta position of the aromatic ring.
  - no resonance delocalization of their electron is possible
  - This addition only increases (alkoxy) or decreases (amino) the lipophilicity of the molecule.

INTERMEDIATE CHAIN.



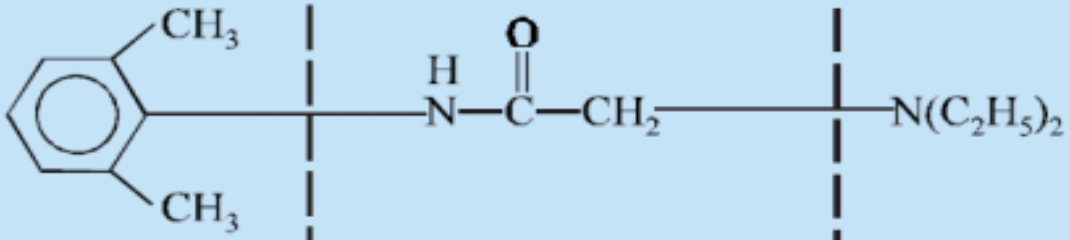
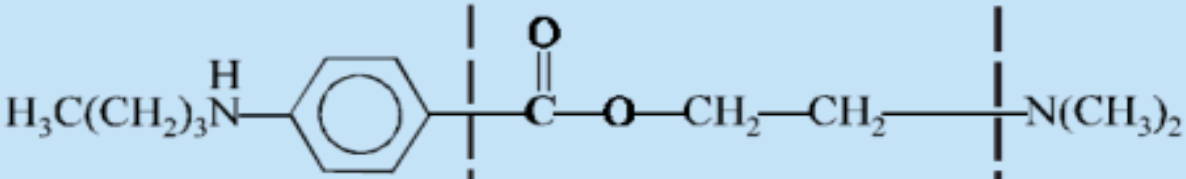
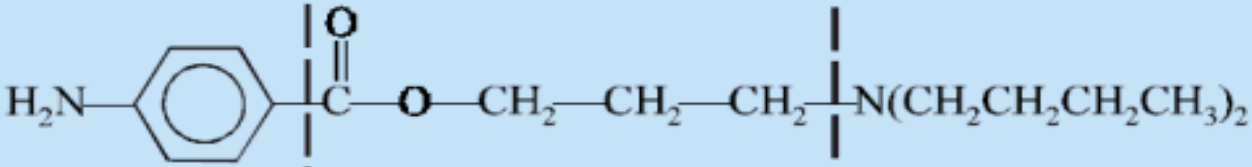
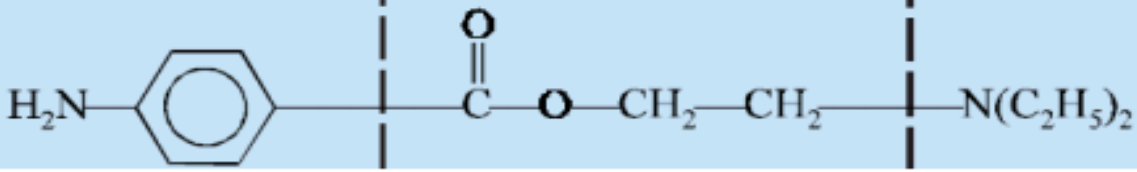
- The intermediate chain almost always contain a short alkylene chain of one to **three carbon** in length.
- The nature of this chain determine the chemical stability of the drug.
- It also influences duration of action and relative toxicity.

HYDROPHILIC CHAIN



- Most clinically useful local anesthetics have a tertiary alkylamine, which readily forms water-soluble salts with mineral acids, and this portion is commonly considered to be the hydrophilic portion of the molecule.
- **The necessity of this portion of the molecule for local anesthetic activity still a matter of debate**
- The strongest opposition comes from the observation that benzocaine, which lack the basic aliphatic amine function, has potent local anesthetic activity.
- the hydrophilic group in most of the clinically useful drugs can be in the form of a secondary or tertiary alkyl.

CONT…

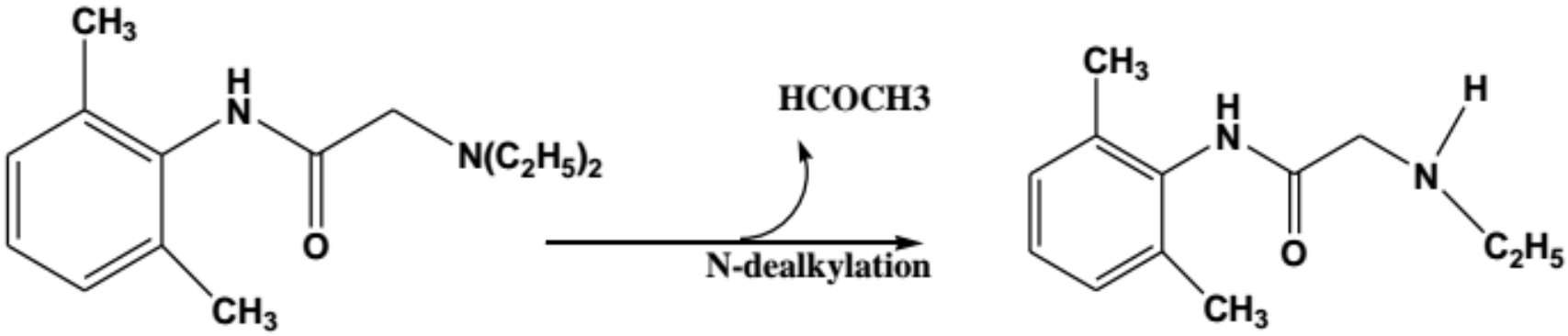
Name	Lipophilic Entity	Intermediate Chain	Hydrophilic Entity
Lidocaine			
Tetracaine			
Butacaine			
Procaine			

Metabolism

- **metabolism** of local anesthetics is important in clinical practice because the overall toxicity of a drug depends not only on its uptake and tissue distribution but also on how it is deactivated in vivo.
- **The amino ester** type local anesthetics are rapidly hydrolyzed by plasma cholinesterase (also known as **pseudocholinesterase**), which is widely distributed in body tissues.
- These compounds can therefore be metabolized in the **blood**, **kidneys**, and **liver** and, to a lesser extent, at the site of administration.

CONT…

- **Ester type anesthetics are hydrolzed by esterase into PABA and the corresponding alcohol.**
  - ✓ It is not surprising that potential drug interaction exists atropine like anticholinergic drugs & sulfonamides.
  - ✓ Amides are biotransformed **in** the liver but **esters** are **hydrolyzed in** the bloodstream by plasma **esterases.**
- **Of all metabolites of lidocaine, only mono ethylgly cinexylidine contribute to some of CNS side effects of lidocaine .**
  - ✓ This suggest the toxicity of lidocaine be related to the removal of the N – ethyl group of lidocaine after crossing BBB



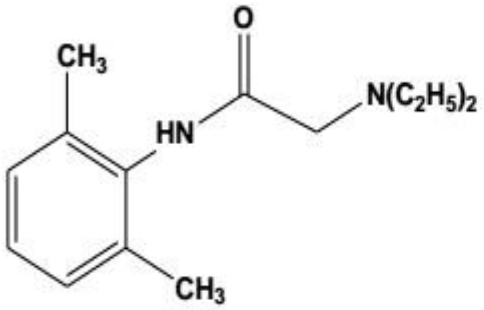
Lidocaine

Monoethylglycinexylidine

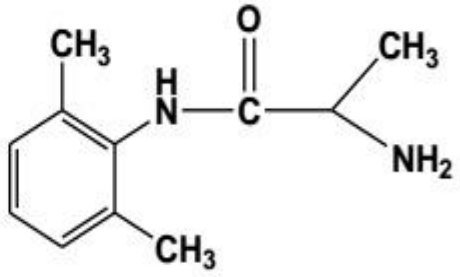
toxic metabolite of lidocaine

CONT…

- To minimze these unwanted side effects of lidocaine; **tocainide** and **tolycaine** have been prepared and found to have no appreciable CNS side effects.
- ✓ **Tocainide** lack vulnerable N – ethyl group has αmethyl group to prevent degradation of primary amine group from amine oxidase.

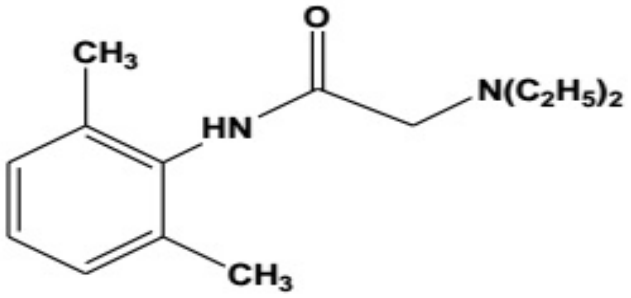


Lidocaine

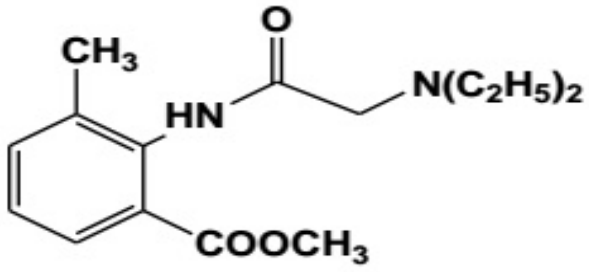


Tocainide

- **Tolycaine** has an o-carbomethoxy substituted for one of o-methyl group of lidocaine.
- ✓ The carbomethoxy group is rapidly hydrolyzed in the blood to polar carboxylic function and is thus unable to cross BBB.



Lidocaine



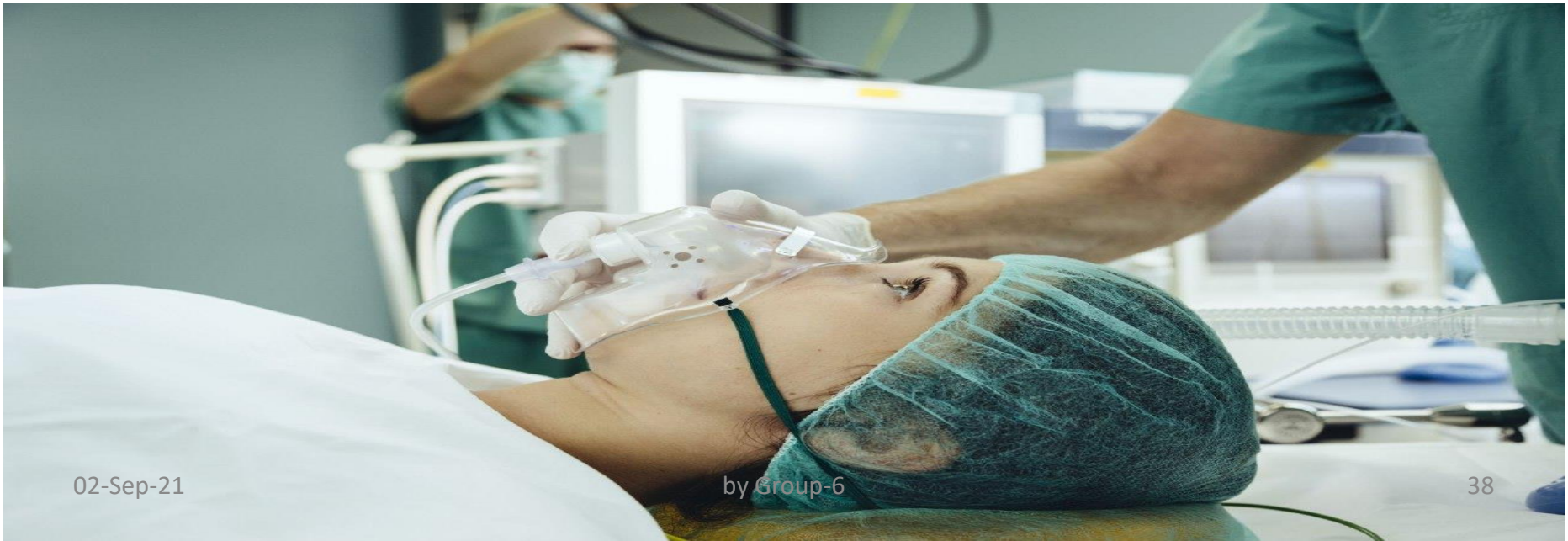
Tolycaine

END OF 2ND PART

WELL COME TO

==Part 3==

GENERAL ANESTHETICS



CONT…

- **General anesthetics** are the drugs, which produce controlled, reversible depression of the functional activities of the central nervous system .
- **Producing loss of sensation and consciousness.**
- Is Anesthesia that affecting the entire body accompanied by loss of consciousness.

CONT…

- GAs are the patients is unconscious and feeling nothing during surgery and patients receive medicine through **Inhaled** and **Iv**

❖ **Inhaled general anesthetics**

- **Ether • Halothane • Desflurane • Enflurane • Isoflurane • Methoxyflurane • Sevoflurane**
- **Nitrous oxide**

❖ **Iv general anesthetics**

• Etomidate • Ketamine • Propofol • Fospropofol • Thiopental

Stages of General Anesthesia

- When an inhalation anesthetic is administered to a patient some of the following well defined stages are produced by increasing the blood concentration of the agent. They are;

1) Stage I (Stage of analgesia)

-This is the period from the beginning of anesthetic administration to the loss of consciousness

-The patient progressively loses pain

Stages of General Anesthesia

2) Stage II (Stage of delirium)

- ✓ This period extends from the loss of consciousness through a stage of irregular and specific breathing to the reestablishment of regular breathing

3) Stage III (Stage of surgical anesthesia)

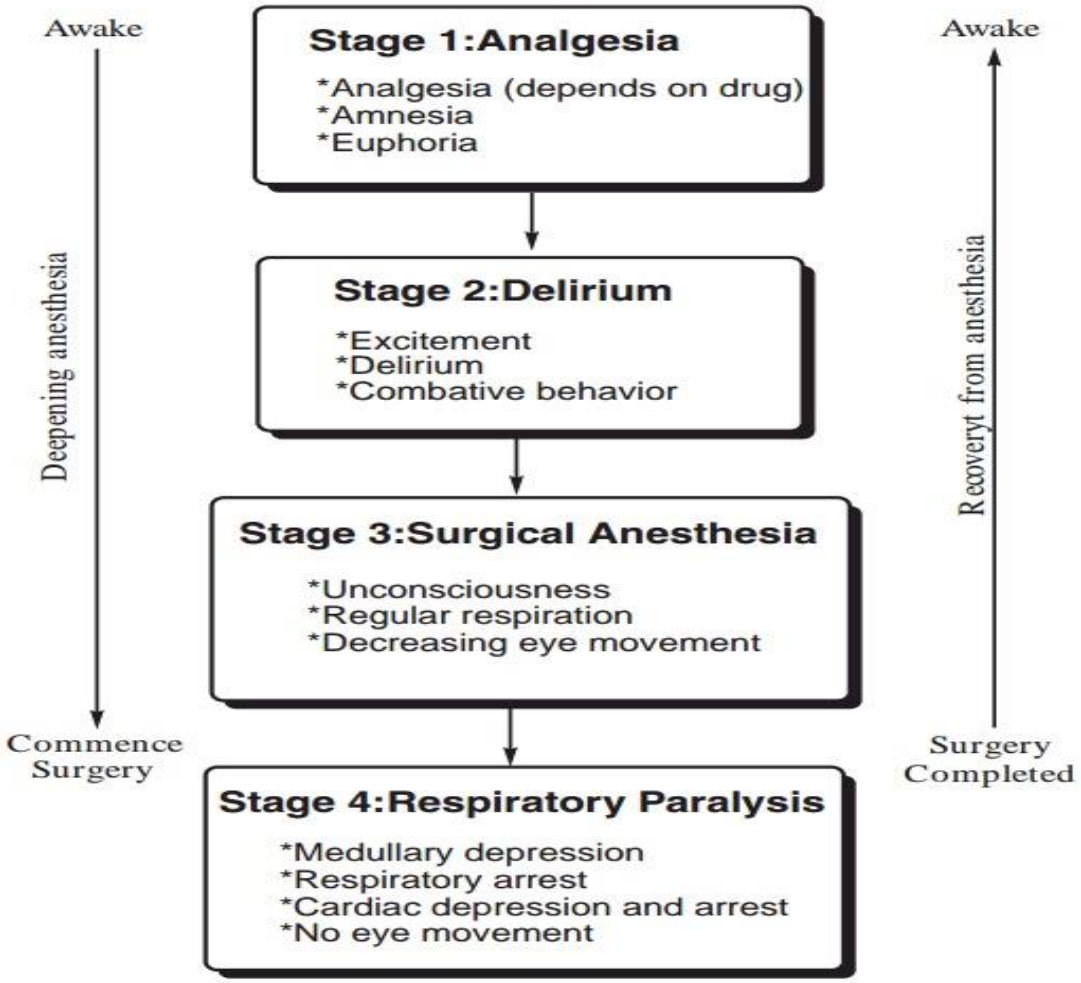
- ✓ In this stage excitement is lost and skeletal muscle relaxation is produced.
- ✓ Most types of surgeries are done in this stage.

CONT…

4) Stage IV (stage of medullary depression)

- ✓ **overdose of the anesthetic may bring the patient to this stage .**
- ✓ **Respiratory and circulatory failure occur in this stage.**
- ✓ **Characterized by respiratory and vasomotor paralysis.**

Summer of Stages of General Anesthesia



Stages of general anesthesia.

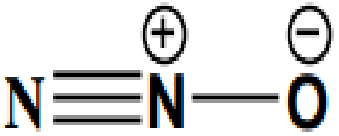
Classification of General Anesthesia

- **General Anesthetics** are classified according to their nature (**volatile** or **non – volatile**) at room temperature. They are:

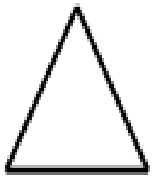
A) Volatile inhalation general anesthetics

- They are administered by inhalation and are further subdivided as;

❖ **Gases:** Nitrous oxide, Cyclopropane, Ethyl chloride



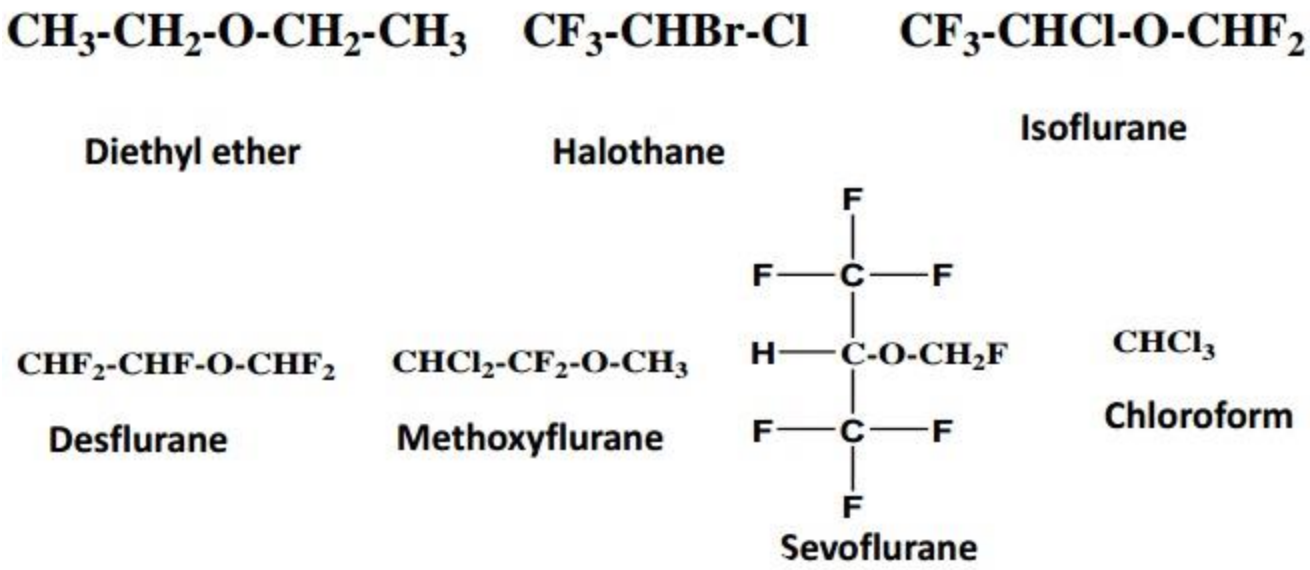
or N_2O



C_2H_5-Cl

CONT…

- **Liquids:** Diethyl ether, Halothane, Chloroform, Trichloroethylene, methoxyflurane, enflurane, isoflurane,desflurane, Sevoflurane



CONT…

B) Non – volatile or intravenous anesthetics

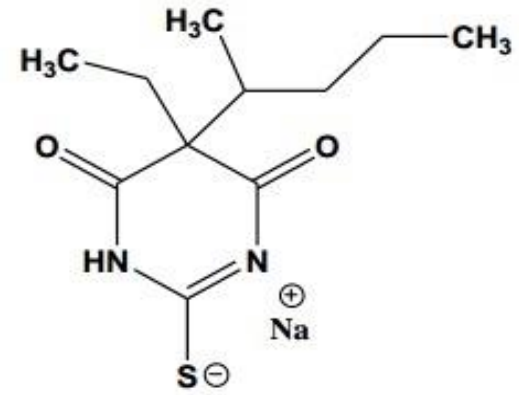
- They are non – volatile at room temperature and are administered by intravenous route.

They are;-

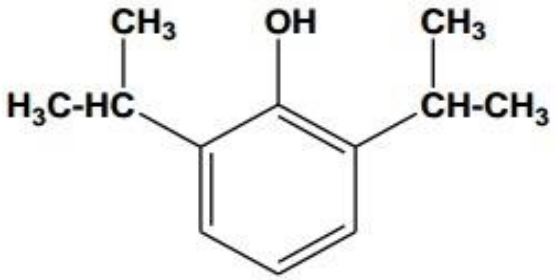
a) Barbiturates: Thiopental sodium, methohexital sodium

b) **Non – barbiturates:** propanidid, propofol.

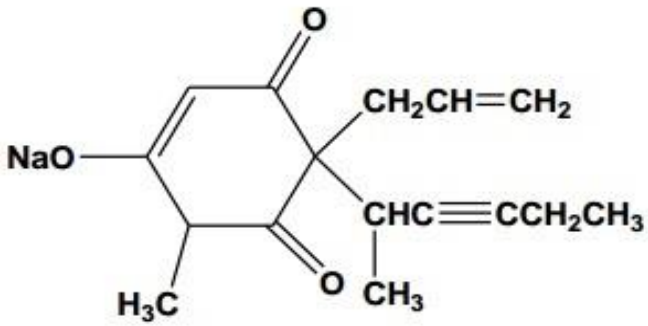
Iv anesthetics



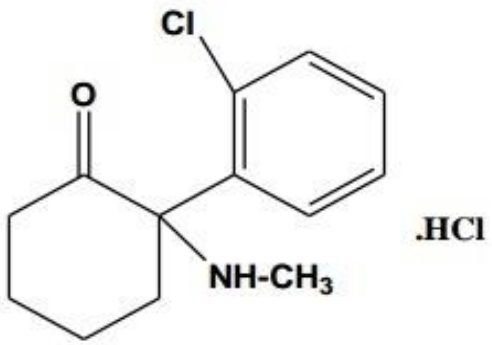
Thiopental sodium



Propofol



Methohexital sodium



Ketamine

Characteristics features of GA

- **Some characteristics of ideal GA**
  - It should be potent
  - It should be non irritating to mucous membrane
  - It should produce rapid and smooth anesthesia
  - **It should produce analgesia and muscle relaxation in addition to anesthesia**
  - It should not produce severe hypotension
  - It should be compatible with adjuvant drugs used in anesthesia mechanism of action

END OF LASTPERESENTATION!!



BYE